

Total No. of Questions :6]

SEAT No. :

P287

[Total No. of Pages :3

Oct./ BE/ Insem. - 605

B.E. (Information Technology)

MACHINE LEARNING AND APPLICATIONS

(2015 Course) (Semester-I) (414454)

Time : 1 Hour]

[Max. Marks :30

Instructions to the candidates:

- 1) Answer Q.NO.1 or 2, Q.NO. 3or 4, Q.NO. 5 or 6.
- 2) Draw neat diagrams wherever necessary.
- 3) Assume suitable data, if necessary.
- 4) Figures to the right indicate full marks.

Q1) a) Write Short note on following two examples of machine learning applications **[5]**

- i) Learning Association
- ii) Reinforcement Learning

b) Explain semi- supervised learning & supervised & unsupervised. **[5]**

OR

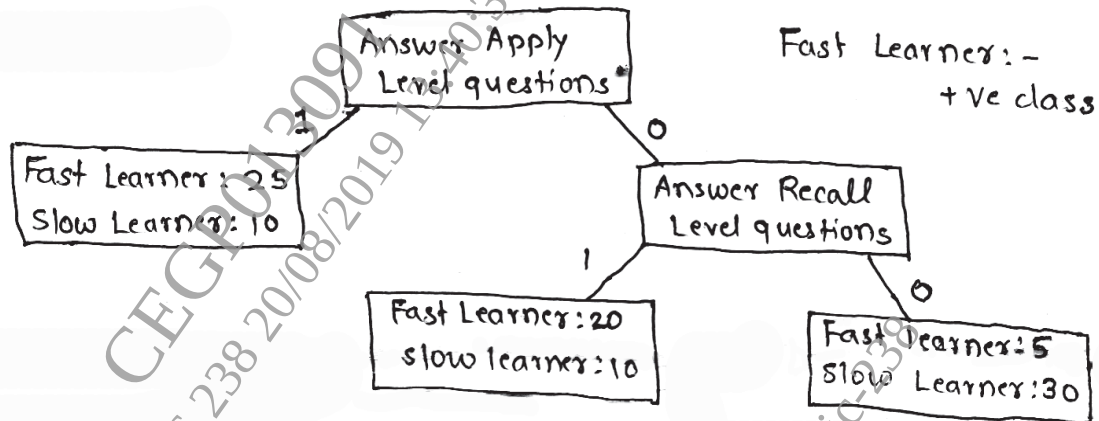
Q2) a) Explain with example forward & backward selection methods of subset selection **[5]**

b) Explain with example K-fold cross validation **[5]**

P.T.O.

Q3) a)

[5]



- Find contingency table.
 - Find Recall.
 - Precision
 - Negative recall.
 - False positive rate.
- b) Write & explain perceptron training algorithm for linear classification [5]

OR

- Q4) a) Consider the three-class confusion matrix. Calculate precision and Recall per [5]

predicted			
Actual	15	2	3
	7	15	8
	2	3	45

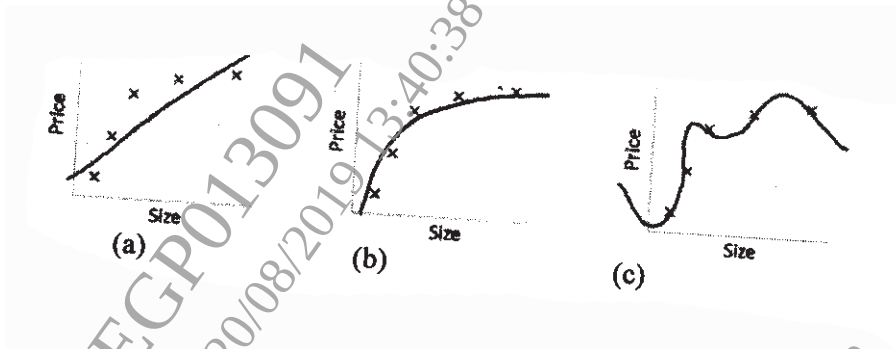
- b) Write short note on Gram matrix and explain with example. [5]

Q5) a) Explain bias-variance dilemma.

[5]

b)

[5]



Explain the above figures (a), (b) and (c)

OR

Q6) a) Justify the following

[5]

- Predict the height of a person. Is it a regression task?
- Find the gender of a person by analyzing his writing style. Is it a classification task?
- Filter out spam emails. Is it an example of unsupervised learning?

b) Suppose you have been given a set of training examples

[5]

$\{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$. Find the equation of the line that best fits the data in that minimizes the squared error.

